to all but a slender view of her abilities. Similarly, since the triumph of string theory is its natural incorporation of quantum mechanics and gravity, and since gravity is bound up with the form of space and time, we should not constrain the theory by forcing it to operate within an already existing spacetime framework. Rather, just as we should allow our artist to work from a blank canvas, we should allow string theory to create its own spacetime arena by starting in a spaceless and timeless configuration.

The hope is that from this blank slate starting point—possibly in an era that existed before the big bang or the pre-big bang (if we can use temporal terms, for lack of any other linguistic framework)—the theory will describe a universe that evolves to a form in which a background of coherent string vibrations emerges, yielding the conventional notions of space and time. Such a framework, if realized, would show that space, time, and, by association, dimension are not essential defining elements of the universe. Rather, they are convenient notions that emerge from a more basic, atavistic, and primary state.

Already, cutting-edge research on aspects of M-theory, spearheaded by Stephen Shenker, Edward Witten, Tom Banks, Willy Fischler, Leonard Susskind, and others too numerous to name, has shown that something known as a zero-brane—possibly the most fundamental ingredient in M-theory, an object that behaves somewhat like a point particle at large distances but has drastically different properties at short ones—may give us a glimpse of the spaceless and timeless realm. Their work has revealed that whereas strings show us that conventional notions of space cease to have relevance below the Planck scale, the zero-branes give essentially the same conclusion but also provide a tiny window on the new unconventional framework that takes over. Studies with these zero-branes indicate that ordinary geometry is replaced by something known as noncommutative geometry, an area of mathematics developed in large part by the French mathematician Alain Connes.4 In this geometrical framework, the conventional notions of space and of distance between points melt away, leaving us in a vastly different conceptual landscape. Nevertheless, as we focus our attention on scales larger than the Planck length, physicists have shown that our conventional notion of space does re-emerge. It is likely that the framework of noncommutative